

Mechanism-labelled clinical evidence with finite-sample error bounds from two-channel saturation genome editing

A method for converting the two outputs of a saturation genome editing screen—cell survival and mRNA abundance—into ACMG/AMP functional-evidence codes (PS3, BS3, and PVS1_RNA). Each code includes a mechanism label, a finite-sample error bound that does not assume a score distribution, and an adjustment for transfer across tissues.

Abstract

Functional assays can provide strong evidence for clinical variant classification. Saturation genome editing (SGE) extends this approach by measuring nearly every possible variant across a target region. Newer SGE screens often report two measurements for each variant: a survival or function score and an mRNA-abundance score. Most current interpretation pipelines use only the survival score. Here, we show that the mRNA measurement adds information that is important for clinical use because it helps identify the mechanism of loss of function. A variant that mainly damages the protein is expected to behave similarly across tissues. A variant that disrupts RNA processing or transcript abundance may be more tissue-dependent and may therefore transfer less directly from the screening cell line to a patient.

We developed a method that converts both measurements into one clinical evidence code for each variant. The code has two parts: the statistical strength of the functional call and the expected transfer of that call across tissues. The method addresses three limitations of current practice. First, we replace the single ClinGen OddsPath estimate with a **class-stratified finite-sample error bound** that makes no assumption about the score distribution. Each issued code therefore has an explicit upper limit on its error rate, and the method abstains when the controls do not support a tier. Second, we use a **Wilks order-statistic tolerance bound** to certify the threshold for an mRNA-abundance drop. This provides a controlled false-positive rate even though the threshold is selected from the data. Third, we apply a **mechanism-aware tissue-transfer adjustment**. RNA-route losses are reduced by one ACMG tier unless independent evidence supports transfer to disease-relevant tissue. The size of this adjustment is calibrated using reproducibility data, and AlphaGenome is used only in regions where its measured performance is adequate.

We applied the method to five published two-channel screens covering 35,333 variants in *BARD1*, *BRCA1*, *PALB2*, *RAD51D*, and *VHL*. The method produces 11 error-bounded ACMG codes, assigns evidence to **4,569** variants that would otherwise remain ungraded, and includes **2,820** protein-blind splice-region variants. It also reduces the transferable tier for **611** RNA-route losses. The error bound achieves 0.958 coverage on held-out controls, whereas a deliberate cross-gene pooling analysis gives lower coverage (0.833), supporting the gene-specific design. The survival component agrees with investigator calls in two external screens at rates of 79–100%. In an additional application, the RNA channel identifies experimentally confirmed splice-altering variants missed by SpliceAI. Using the Wilks-certified RNA threshold increases precision from 0.84 to 0.89 without reducing recovery.

1. Introduction

1.1 Variant interpretation remains a clinical bottleneck

A human genome contains millions of variants, and clinical sequencing can now identify variants in medically important genes within hours. The more difficult step is deciding what those variants mean. Under the ACMG/AMP framework, many rare variants are reported as **variants of uncertain significance (VUS)**. These variants are neither clearly actionable nor clearly benign. For example, a VUS in a cancer-predisposition gene such as *BRCA1* or *PALB2* may not provide enough evidence for risk-reducing surgery, testing of relatives, or use of a mechanism-specific therapy.

Reducing the number of VUS is therefore a major goal in clinical genomics. The problem is especially difficult for splice-region, deep-intronic, and other non-coding variants. Their effects are often hard to infer from DNA sequence alone, and many have little or no clinical observation data. These are also the variant classes for which a direct functional measurement may be most useful.

1.2 Functional assays provide strong evidence, and SGE can measure variants at scale

The ACMG/AMP framework allows a well-validated functional assay to support either pathogenicity or benignity. These evidence types are represented by **PS3** for functional evidence supporting pathogenicity and **BS3** for functional evidence supporting a benign interpretation. ClinGen commonly assigns the strength of this evidence—Supporting, Moderate, or Strong—using **OddsPath**. OddsPath measures how well an assay separates known pathogenic controls from known benign controls and maps the resulting odds to a fixed evidence ladder.

Saturation genome editing (SGE) is a particularly broad source of functional evidence. SGE introduces many or all possible single-nucleotide variants into a genomic region in its native context and then measures their effect on cell fitness or gene function. The assay can therefore generate data for variants that have never been observed in a patient. It changes functional testing from a one-variant experiment into a regional resource.

As SGE datasets become larger, the main challenge is no longer only generating the data. The next challenge is converting those measurements into evidence that is statistically calibrated, clinically interpretable, and appropriate for use outside the cell system in which the screen was performed. This study focuses on that conversion.

1.3 The second channel provides information about mechanism

Recent SGE studies report **two measurements for each variant**. In addition to a survival or function score, they measure the **mRNA abundance** of the edited allele, often by capturing the transcript and comparing its representation with the input library. The two measurements answer related but different questions. The survival score asks whether the variant reduces gene function. The mRNA score can help explain how that loss occurs.

A variant may reduce function while leaving transcript abundance largely unchanged. Such a result is consistent with a protein-level effect, such as altered protein structure or activity. A different variant may reduce both function and mRNA abundance. That pattern is consistent with an RNA-level effect, including disrupted splicing, reduced transcript stability, or nonsense-mediated decay. The mRNA channel does not identify every molecular event on its own, but it provides direct evidence that is absent from a survival-only analysis.

This distinction matters because the mechanism affects transfer across tissues. A direct protein defect is generally expected to be less sensitive to tissue-specific RNA processing. By contrast, splicing and transcript-abundance effects can vary between cell types. An exon that is skipped in the screening cell line may be retained in the disease-relevant tissue, or the opposite may occur. A pipeline that reads only the survival score

treats these mechanisms as equivalent and loses information needed to judge how strongly the assay result should transfer to a patient.

1.4 Three limitations in current weighting of functional evidence

The use of only one channel is part of a broader problem. Current methods for converting functional assays into clinical evidence have three important limitations.

1. **OddsPath is usually reported as a point estimate.** The estimate is calculated from a finite set of controls, often only a few dozen within a gene and variant class. A point estimate does not show how high the true error rate could be at that sample size. A code such as PS3_Strong may therefore appear more certain than the supporting controls justify. Because functional evidence can affect the final clinical classification, overstating its strength can have direct clinical consequences.
2. **The evidence strength does not usually include mechanism or tissue transfer.** A survival-only analysis gives the same functional interpretation to a protein-mediated loss and an RNA-mediated loss when their scores are similar. However, the RNA-mediated result may be more dependent on the cell type used in the assay. A mechanism-blind method can therefore give too much transferable weight to some RNA-route variants.
3. **Protein-focused interpretation provides no clear route for protein-blind variants.** Splice-region and intronic variants may alter function without changing the encoded amino acid in a way that a protein-centred analysis can use. These variants are common among VUS and are often the variants for which the mRNA channel is most informative. A calibrated functional code for these classes is still lacking.

1.5 Overview of the study

We present a method that converts the two SGE measurements into a **mechanism-labelled and transfer-adjusted ACMG/AMP evidence code with a finite-sample error bound**. The procedure is designed to address the three limitations above. It is evaluated using five published two-channel SGE screens containing 35,333 variants. The numerical results are generated from fixed inputs, and the statistical claims are tested on held-out or external data where possible.

Section 2 tests whether the mRNA channel separates protein-route and RNA-route losses. Section 3 describes the statistical bounds, the RNA threshold, and the tissue-transfer rule. Section 4 reports the evidence codes, the number and type of variants affected, and the validation analyses. Sections 5 and 6 describe the scope of the claims and the reproducibility workflow. Tables 1–14 are provided at the end of the manuscript.

2. The mechanism signal can be separated

The method assumes that the mRNA channel helps distinguish protein-route loss from RNA-route loss. We first tested whether this signal is present in the five screens.

We combined the PASS-filtered and VEP-annotated variants from all five studies. Each variant was assigned to a shared three-class outcome—**normal, uncertain, or loss of function**—using the survival score and the thresholds reported by the original study. This produced 27,793 normal variants, 5,497 loss-of-function variants, and 2,043 uncertain variants among 35,333 total variants (Table 1). We then examined the loss-of-function variants and asked how often the mRNA channel showed a transcript drop. Variants were grouped by whether their annotated consequence could directly damage the protein.

The difference was large and was present in all five genes (Figure 1a and Table 2). Among **protein-blind** loss-of-function variants—synonymous or splice-region variants that do not directly encode a damaging amino-acid change—**74.2%** showed an mRNA drop. The corresponding value for missense loss-of-function variants

was **26.6%**, giving a pooled odds ratio of **7.9**. When a variant cannot readily explain the functional loss through a coding change but produces both loss of function and lower mRNA abundance, an RNA mechanism is the most direct explanation. This pattern is especially useful for splice-region variants, for which the RNA effect is measured rather than inferred only from sequence.

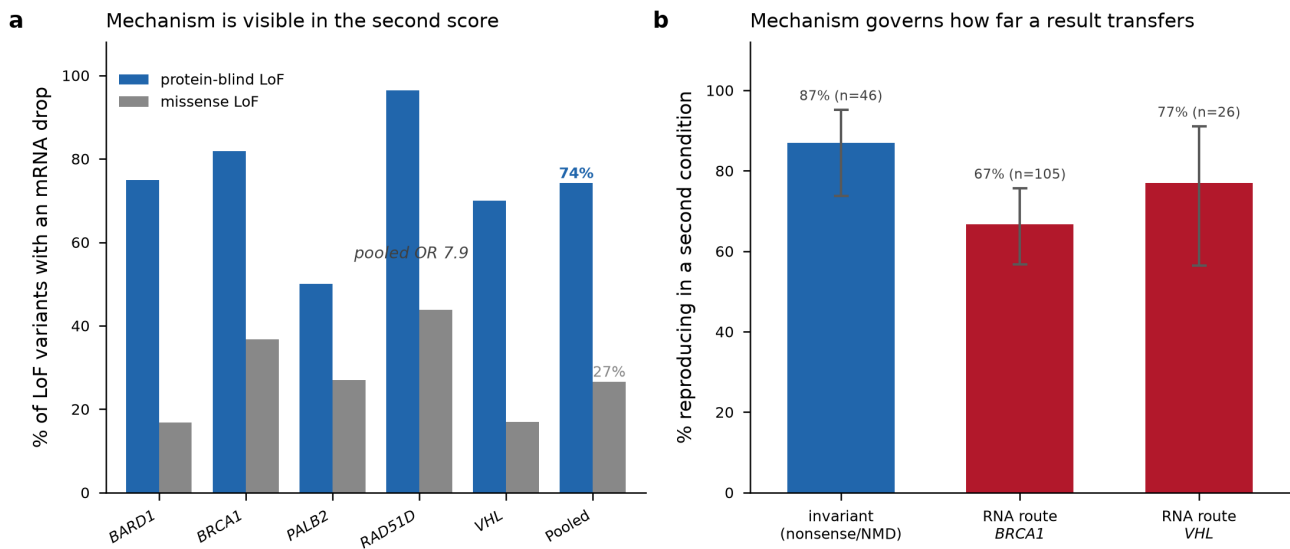


Figure 1. The second channel helps identify mechanism, and mechanism affects transfer across conditions. (a) Among loss-of-function variants, protein-blind variants (synonymous and splice-region) show an mRNA-abundance drop more often than missense variants (74.2% versus 26.6% in the pooled analysis; odds ratio 7.9). A protein-blind functional loss with reduced mRNA is consistent with an RNA-mediated mechanism. (b) An mRNA effect caused by a largely tissue-invariant mechanism, nonsense-mediated decay, reproduces in an independent measurement condition in 87% of cases. Tissue-variable RNA-route effects reproduce in 67% of BRCA1 cases and 77% of VHL cases. This difference is used to calibrate the tissue-transfer adjustment.

Two additional findings affect the design of the method.

The two channels are positively correlated. Within RNA-route losses, error in the survival score and error in the mRNA score are not independent. The observed association ranges from approximately $\phi = +0.23$ in *PALB2* to $+0.62$ in *RAD51D*. If the two error bounds were multiplied as though the channels were independent, the combined error would be underestimated by about eight- to nine-fold (Section 3.4). The joint error must therefore be bounded directly.

Some variants show evidence for two mechanisms. We identified **501 missense loss-of-function variants** that also had a measured mRNA drop: 155 in *BRCA1*, 110 in *PALB2*, 100 in *BARD1*, 96 in *RAD51D*, and 40 in *VHL*. These variants may affect the protein while also producing an RNA-level consequence. A one-channel analysis cannot represent this pattern well, so the method handles this group separately when applying the tissue-transfer rule (Section 3.5).

The data also support a link between mechanism and reproducibility across conditions (Figure 1b). An mRNA effect associated with nonsense-mediated decay reproduced in an independent condition in **87%** of cases. RNA-route effects expected to be more tissue-variable reproduced in **67–77%** of cases. This observed gap provides the empirical basis for the one-tier transfer adjustment described in Section 3.5 and evaluated in Section 4.7.

3. Method

For each variant, the method combines the survival and mRNA scores into one evidence code. The code has two axes. The first axis represents the statistical confidence in the functional call. The second represents how

strongly the result is expected to transfer from the screening system to disease-relevant tissue. Three components distinguish the method from a standard OddsPath analysis.

3.1 The three-class assay call does not use clinical labels

The survival score is first converted into the shared three-class call using the thresholds defined by the original investigators. ClinVar and gnomAD labels are not used to assign this call. The call is therefore based on the assay result rather than on the later clinical truth set.

This separation is important because the same clinical labels are used later to estimate error. If labels were used both to define the assay classes and to evaluate them, information from the controls could leak into the calls being graded. Keeping the steps separate avoids that circularity. Despite being label-free, the survival call agrees with 96.3% of pathogenic and 94.5% of benign ClinVar controls with at least two review stars. Agreement remains similar in the held-out controls: 96.5% for pathogenic variants and 93.9% for benign variants (Table 3). These results indicate that the call is not simply fitted to the calibration controls.

3.2 Component 1: class-stratified finite-sample error bounds

Instead of using only an OddsPath point estimate, the method places an upper bound on the rate at which the survival call is wrong within the relevant variant class. Two errors are considered:

- the **false-loss rate**, a benign control called loss of function, which limits a PS3 claim; and
- the **false-normal rate**, a pathogenic control called normal, which limits a BS3 claim.

Each rate receives an **exact Clopper–Pearson 95% upper confidence limit**. Because each control is classified as correct or incorrect, an exact binomial bound is appropriate and does not require an assumed distribution for the continuous assay score.

Variants at the same genomic position may share local sequence or assay effects. The analysis therefore uses a **position-clustered effective sample**, so variants at one position are not counted as fully independent trials. Three correlation models gave upper limits within 0.007 of one another; the position-clustered result is used throughout (Table 4).

Two rules preserve the meaning of the bound. First, variant classes remain separate: false loss is estimated from synonymous negative controls, and false normal from nonsense positive controls. Second, genes are not pooled, because Section 4.6 shows that error rates differ between genes.

The upper error limit is converted to a lower OddsPath bound and compared with the standard thresholds: Supporting at 2.1, Moderate at 4.3, and Strong at 18.7. A code is issued only when the bounded odds reach a threshold; otherwise, the method abstains. This produces conservative tiers that reflect both the observed errors and the number of available controls.

3.3 Component 2: a Wilks-certified threshold for an mRNA drop

The RNA channel requires a threshold below which an mRNA-abundance score counts as a drop. A natural starting point is the fifth percentile of functionally normal synonymous controls within each gene, corresponding to a nominal 5% false-positive rate. However, this threshold is selected from the same controls used to report its apparent error, so the observed rate is optimistic when the sample is finite.

We use a **Wilks order-statistic tolerance bound** to account for this selection. For an empirical quantile from n control values, the probability mass beyond the selected order statistic has an exact $\text{Beta}(k, n-k+1)$ distribution under any continuous score distribution. The 95% upper tolerance limit therefore bounds the true false-positive rate without a parametric model for the mRNA scores.

The certified threshold is the most permissive observed cut whose Wilks upper limit remains at or below 5%. It is more negative, and therefore stricter, than the preliminary cut; the difference is larger when fewer controls are available (Table 6). For *VHL*, the threshold changes from -0.597 , with an upper limit of 0.071 , to -0.813 ,

with an upper limit of 0.047. For *BARD1*, it changes from -0.684 (0.060) to -0.759 (0.049). Section 4.9 shows that the certified threshold improves independent precision without reducing recovery.

3.4 Joint error across two correlated channels

An RNA-route call depends on both channels: the survival score indicates loss of function, and the mRNA score indicates an RNA-level effect. The error for this joint call cannot be obtained by multiplying the separate channel bounds because the two measurements are positively correlated (Section 2). Multiplication would treat the errors as independent and would underestimate the observed joint error by about eight- to nine-fold.

The method therefore calculates a **combined two-channel false-positive upper bound** directly within each gene. It uses the smallest bound that remains valid under the observed dependence, rather than the product of the two marginal bounds. The resulting gene-specific upper limits range from 0.013 to 0.065. At no stage are the two error probabilities multiplied under an independence assumption.

3.5 Component 3: mechanism-aware adjustment for tissue transfer

The mRNA result and the annotated molecular consequence are used to assign a mechanism route.

- **Protein route:** the loss is attributed mainly to the protein, as for a missense or in-frame indel, or to a nonsense variant with an mRNA drop consistent with constitutive nonsense-mediated decay. These effects are treated as relatively tissue-invariant.
- **RNA route:** the loss is attributed to altered splicing or transcript abundance and may vary between tissues.

A protein-route loss keeps the tier assigned by the first axis. An RNA-route loss is reduced by one ACMG tier—Strong to Moderate, Moderate to Supporting, or Supporting to uncertain—unless independent evidence supports transfer. Dual-mechanism variants are treated as RNA route for this adjustment because their RNA consequence remains tissue-sensitive.

Corroboration must be independent. An RNA-route loss keeps its tier if the mRNA drop reproduces in an independent condition from the same study, or if AlphaGenome predicts a splice disruption with concordant disease-tissue expression. The original mRNA measurement does not count as corroboration because it is the observation being transferred.

AlphaGenome is used only where it performs adequately. Its splice scores were compared with the measured mRNA outcome and grouped by distance from the nearest junction (Table 7). AUROC is 0.79 within 3 nucleotides of a junction but about 0.60 in the exon core. AlphaGenome can therefore corroborate only strata whose lower AUROC confidence limit exceeds 0.70. This gate includes **2,656 of 12,904** intronic tissue-bridge variants.

The one-tier change is calibrated. The adjustment is based on the observed difference in reproduction rates between nonsense-mediated decay and tissue-variable RNA-route effects. Section 4.7 maps that difference to the OddsPath scale.

3.6 Final output

Each held-out variant receives one mechanism-labelled and transfer-adjusted ACMG/AMP result. Protein-route calls receive PS3 or BS3 at the supported tier. RNA-route losses receive PVS1_RNA at the transfer-adjusted tier. Variants for which the controls do not support a tier are reported as **uncertain**. The complete pipeline starts from fixed input files and can be run with one command.

4. Results

4.1 Data and fixed calibration and held-out sets

The analysis includes five published two-channel SGE screens (Table 1). Variants were matched to ClinVar using GRCh38, release 2026-07-06, with MD5 verification, and to gnomAD r4. ClinVar records with at least two review stars were used as controls (Table 3).

A fixed seeded split (seed 20260710) created a **calibration set** of 2,927 controls and a separate **held-out set** of 32,406 variants. The calibration controls were further divided into fit and validation subsets of 1,461 and 1,466 variants for the coverage test. No calibration variant was used to assign a held-out code. Fixed inputs and the seeded split reproduce the reported numbers exactly; this computational check is separate from the held-out and external validation below.

4.2 The method supports 11 error-bounded ACMG codes

The class-specific bounds support **11 ACMG/AMP functional-evidence codes** across the five genes: 8 Moderate and 3 Supporting (Table 5). Their upper error limits are shown in Table 4.

The method abstains when the limit is too high. For example, nonsense variants in *RAD51D* and *VHL* have false-normal upper limits of 0.17 and 0.40, so those strata receive no code. By contrast, splice-region strata receive a code in all five genes. These protein-blind variants are difficult to grade with a protein-only interpretation, but the two-channel approach provides a calibrated route.

4.3 Result 1: 4,569 previously ungraded variants receive evidence codes

When applied to the 32,406 held-out variants, the method gives a transferable evidence code to **30,133** variants. It assigns a code to **4,569** variants that would remain uncertain in a mechanism-blind, protein-centred analysis.

Among these newly graded variants, **2,820 are protein-blind splice-region variants**. This is the group highlighted in Section 1.4 because a direct protein interpretation cannot explain their functional loss. The final RNA-route code set contains 804 PVS1_RNA assignments, all involving splice-region or intronic consequences (Table 13). These results show that the second channel is most useful in the variant classes for which clinical interpretation is often most difficult.

4.4 Result 2: the mRNA channel reduces the tier for 611 RNA-route losses

Including the mRNA channel changes the transferable ACMG tier for **611** held-out variants compared with the same pipeline without the mechanism component (Table 11). Every change is a reduction for an RNA-route loss: 207 in *PALB2*, 172 in *BARD1*, 146 in *RAD51D*, 64 in *BRCA1*, and 22 in *VHL*. This within-pipeline comparison isolates the effect of the second channel; a survival-only analysis would have assigned these variants a stronger transferable tier.

Among 421 held-out dual-mechanism variants, 345 are reduced and 76 retain their tier because they have independent corroboration. The rule therefore does not reduce every variant with an mRNA effect.

4.5 Validation: the finite-sample bound covers held-out errors

For each gene and class, the error upper limit was estimated in the fit subset and compared with the observed error in the separate validation subset. Twenty-four strata had enough data for this test. The upper limit covered the held-out error in **23 of 24 strata**, giving empirical coverage of **0.958** (Table 8). This is consistent with the target coverage of 0.95.

Agreement between the assay call and the clinical labels in the held-out controls was 0.965 for pathogenic variants called loss of function and 0.939 for benign variants called normal. These controls were not used to

set the error limit being tested. The result therefore supports the finite-sample procedure on data not used for fitting.

4.6 Validation: cross-gene pooling reduces coverage

The main method estimates each error bound within one gene. To test whether this restriction is necessary, we performed a leave-one-gene-out analysis that deliberately estimated a gene's bound from controls in the other genes. Coverage fell to **0.833**, or 25 of 30 tested strata, which is below the 0.95 target (Table 8).

Adding more donor genes did not correct the problem. Coverage changed from approximately 0.90 with one donor gene to 0.83 with four donor genes (Table 9). This negative-control analysis indicates that the relevant error rates are gene-specific. A larger pooled control set can appear more precise while still giving the wrong coverage for the target gene. The result supports the decision not to borrow controls across genes.

4.7 Validation: the one-tier tissue adjustment is consistent with reproducibility data

The tissue-transfer rule reduces RNA-route evidence by one ACMG tier. A nonsense-mediated-decay effect reproduced in an independent condition in **87.0%** of cases, compared with **66.7%** for *BRCA1* RNA-route effects and **76.9%** for *VHL* RNA-route effects (Figure 1b and Table 10).

The direct difference corresponds to an odds ratio of 3.33, or **1.68 OddsPath tiers**, with a 95% confidence interval from -0.13 to 3.74 tiers. The interval is wide but includes the chosen one-tier reduction. These are repeated measurements within the same cell lineage, not patient-tissue comparisons, so the result supports a cautious adjustment rather than direct proof of cross-tissue transfer.

4.8 Validation: the survival component transfers to external screens

We next evaluated the survival component using two independent one-channel SGE studies that were not used to develop the method: **RAD51C** from Olvera-León *et al.* in *Cell* (2024) and **DDX3X** in an X-linked neurodevelopmental disorder study. Because these screens do not include an mRNA channel, this analysis tests only the first axis.

The survival call agreed with independent ClinVar controls in both datasets (Table 12). For *RAD51C*, 98.2% of benign controls were called normal and 100% of pathogenic controls were called loss of function. For *DDX3X*, the corresponding values were 100% and 79.4%. The lower pathogenic agreement in *DDX3X* indicates that performance is not uniform across genes, but the overall results show that the survival procedure is not restricted to the five genes used for development.

4.9 Result 3: the RNA channel identifies splice effects missed by SpliceAI

We compared the measured RNA signal with **SpliceAI**, a widely used sequence-based splice predictor. Among missense and synonymous variants, the initial RNA threshold identified **1,614 candidates** with reduced transcript abundance.

Independent evidence came from **SpliceVarDB**, which includes minigene, RT-PCR, and RNA-sequencing results. Entries citing one of the five source SGE screens were excluded.

Precision. Thirty-three candidates had independent SpliceVarDB data, and 16 were confirmed to alter splicing. Conservative precision was **0.84** and inclusive precision was 0.91 (Table 14). The Wilks-certified threshold increased conservative precision to **0.89** and inclusive precision to 0.92 while retaining all 16 confirmed variants (Figure 2a).

Variants missed by SpliceAI. Of these 16 confirmed variants, SpliceAI missed 1 at a Δ -score threshold of 0.2, **7 of 16** at 0.5, and **14 of 16** at 0.8 (Figure 2b). Many lie in the exon core, more than 3 nucleotides from a splice junction, where the measured mRNA signal adds information not captured by sequence prediction.

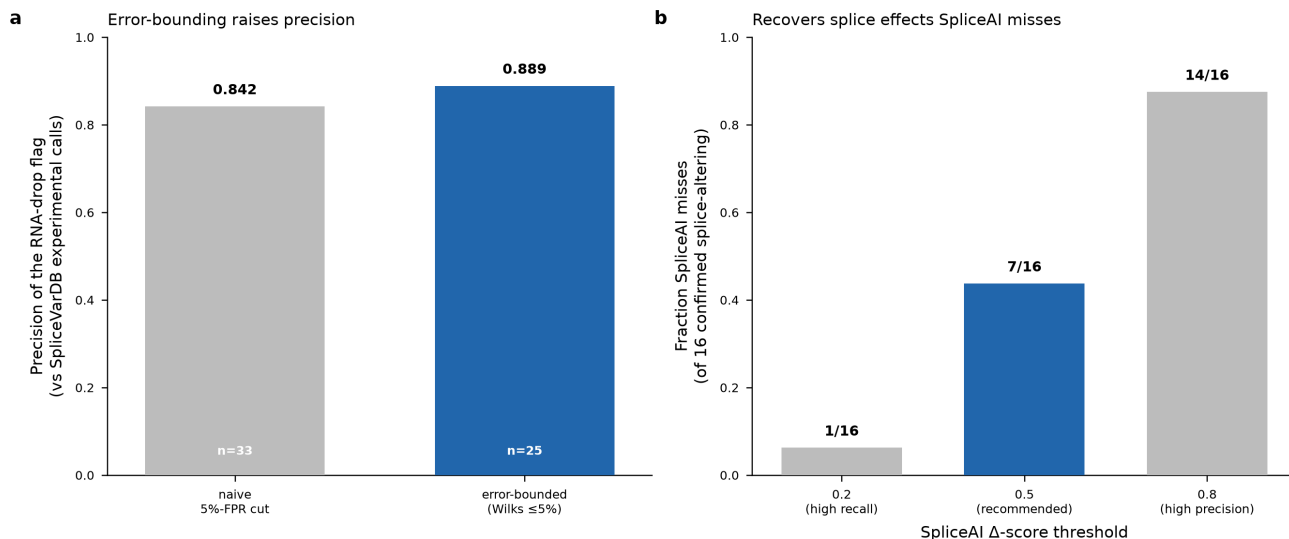


Figure 2. The measured RNA channel identifies experimentally confirmed splice effects that SpliceAI misses. (a) Against independent SpliceVarDB results, the Wilks-certified threshold increases conservative precision from 0.84 to 0.89 without reducing recovery. (b) Among 16 confirmed variants detected by the RNA channel, SpliceAI classifies 1, 7, and 14 as negative at Δ -score thresholds of 0.2, 0.5, and 0.8, respectively.

The methods are complementary. SpliceAI provides genome-wide sequence-based evidence, whereas the RNA channel directly measures transcript effects in the screened region. The comparison identifies a clinically relevant group of splice effects that sequence prediction alone misses at commonly used thresholds.

5. Scope and limitations

The transfer adjustment is not a patient-tissue measurement. It is calibrated using a second condition in the same experimental system. Direct RNA evidence from disease-relevant patient tissue remains stronger.

Evidence strength comes from the survival component. Agreement between survival loss and an mRNA drop is used to assign mechanism and transfer, not to increase the original tier.

External validation covers only one axis. The *RAD51C* and *DDX3X* screens lack an mRNA channel, so they test only survival. The mechanism component is evaluated through within-study reproduction and the independent SpliceVarDB comparison.

Control counts limit some codes. Small gene-by-class control sets produce wider error bounds and more abstention. This is an intended feature of the finite-sample approach.

6. Reproducibility

```
python run.py          # error bounds → strength → mechanism → evidence codes
python run.py --all    # also rebuild clinical controls from ClinVar/gnomAD
python run.py --validate # also run the validation tests
```

The main output is one mechanism-labelled and transfer-adjusted ACMG/AMP code for each held-out variant. All tables are regenerated under `results/`, and both figures are stored in `manuscript/figures/`. The fixed inputs and seeded split allow the reported numerical results to be reproduced exactly.

Tables

All tables are regenerated from `results/` by `python run.py`. The values below are the fixed results used in this manuscript, and the tables follow the order in which they are discussed.

Table 1. Variant counts and shared three-class survival calls for the five two-channel SGE screens.

Gene	Variants	Normal	Uncertain	LoF
BARD1	10911	8866	230	1815
BRCA1	3893	2823	249	821
PALB2	12851	9955	1287	1609
RAD51D	5410	4383	134	893
VHL	2268	1766	143	359
Total	35333	27793	2043	5497

Table 2. Protein-blind loss-of-function variants show an mRNA drop more often than missense loss-of-function variants (74.2% versus 26.6% in the pooled data; odds ratio 7.9).

Gene	Protein-blind LoF (n)	% with mRNA drop	Missense LoF (n)	% with mRNA drop
BARD1	12	75.0	597	16.8
PALB2	28	50.0	407	27.0
RAD51D	28	96.4	219	43.8
BRCA1	11	81.8	422	36.7
VHL	10	70.0	237	16.9
POOLED	89	74.2	1882	26.6

Table 3. Fixed clinical controls from ClinVar (at least two review stars) and gnomAD r4, using seed 20260710. On held-out controls, pooled agreement is 96.5% for pathogenic variants called LoF and 93.9% for benign variants called normal.

Gene	Benign controls	Pathogenic controls
BARD1	600	131
BRCA1	287	349
PALB2	644	251
RAD51D	298	51
VHL	177	139
Total	2006	921

Table 4. Finite-sample error bounds for the survival call. Synonymous controls are treated as functional negatives, so a LoF call is a false loss. Nonsense controls are treated as functional positives, so a normal call is a false normal. The final value is the position-clustered Clopper–Pearson 95% upper limit.

Gene	Control class	Error type	n	errors	observed rate	eff. clusters	95% ceiling
BARD1	synonymous	false-LoF (benign→LoF)	1347	12	0.0089	754	0.0179
BRCA1	synonymous	false-LoF (benign→LoF)	579	11	0.0190	327	0.0359
PALB2	synonymous	false-LoF (benign→LoF)	1919	29	0.0151	1059	0.0221
RAD51D	synonymous	false-LoF (benign→LoF)	679	28	0.0412	323	0.0657
VHL	synonymous	false-LoF (benign→LoF)	387	14	0.0362	204	0.0583
BARD1	nonsense	false-Normal (nonsense→Normal)	368	3	0.0082	326	0.0236
BRCA1	nonsense	false-Normal (nonsense→Normal)	138	0	0.0000	124	0.0239
PALB2	nonsense	false-Normal (nonsense→Normal)	479	5	0.0104	420	0.0217
RAD51D	nonsense	false-Normal (nonsense→Normal)	95	11	0.1158	89	0.1698
VHL	nonsense	false-Normal (nonsense→Normal)	60	15	0.2500	54	0.3950

Table 5. Eleven ACMG/AMP functional-evidence codes supported by the class-matched Clopper–Pearson bounds and mapped to the OddsPath ladder: 8 Moderate and 3 Supporting.

Gene	Class	Direction	ACMG tier	OddsPath (lower bd)	n path ctrl	n benign ctrl
BARD1	splice_region	BS3	Moderate	8.07	43	60
BARD1	splice_region	PS3	Moderate	13.39	43	60
BRCA1	missense	BS3	Moderate	13.98	142	19
BRCA1	missense	PS3	Supporting	3.56	142	19
BRCA1	splice_region	BS3	Moderate	10.17	120	63
BRCA1	splice_region	PS3	Moderate	11.27	120	63
PALB2	splice_region	BS3	Moderate	6.05	61	47
PALB2	splice_region	PS3	Supporting	4.19	61	47
RAD51D	splice_region	BS3	Moderate	4.78	20	65
RAD51D	splice_region	PS3	Moderate	10.61	20	65
VHL	splice_region	PS3	Supporting	2.23	11	18

Table 6. Wilks-certified thresholds for an mRNA-abundance drop. The preliminary threshold is the empirical fifth percentile of synonymous negative controls. The certified threshold is the most permissive cut whose Wilks 95% upper limit remains below the nominal false-positive rate. Certification makes the threshold stricter, especially when fewer controls are available.

Gene	Nominal FPR	n	Prelim threshold	Prelim point FPR	Prelim Wilks ceiling	Certified threshold	Certified point FPR	Certified Wilks ceiling
BARD1	0.05	1319	−0.684	0.0500	0.0602	−0.759	0.0402	0.0494
PALB2	0.05	1659	−0.564	0.0500	0.0591	−0.604	0.0416	0.0499
RAD51D	0.05	641	−1.016	0.0499	0.0647	−1.177	0.0359	0.0486
BRCA1	0.05	536	−0.667	0.0504	0.0667	−0.835	0.0354	0.0494
VHL	0.05	355	−0.597	0.0507	0.0710	−0.813	0.0310	0.0473

Table 7. AlphaGenome reliability by distance from the nearest splice junction. The predictor is used for corroboration only when the lower AUROC confidence limit is above 0.70. This condition is met near the junction (≤ 3 nucleotides) but not in the exon core. The gate includes 2,656 of 12,904 intronic tissue-bridge variants.

Distance-to-junction stratum	n	mRNA-drop events	AUROC	95% CI	Reliable (lower CI > 0.70)?
(all measurable, overall)	22429	2521	0.627	[0.61, 0.64]	no
exonic_splice_region (≤ 3)	1025	309	0.794	[0.76, 0.82]	yes
exonic_core (> 3)	21404	2212	0.597	[0.58, 0.61]	no

Table 8. Coverage of the finite-sample bound on held-out controls. The within-gene analysis matches the intended use of the method. The leave-one-gene-out analysis deliberately pools genes and serves as a negative control.

Test	Strata tested	Covered	Coverage	Target	Result
Within-gene (method as used)	24	23	0.958	≥ 0.95	PASS
Leave-one-gene-out (negative control)	30	25	0.833	≥ 0.95	by design < target

Table 9. Cross-gene coverage by number of donor genes. Adding donor genes does not restore coverage, supporting gene-specific error estimation.

# donor genes	Coverage	# evaluations
1	0.9038	104
2	0.8538	171
3	0.8305	118
4	0.8333	30

Table 10. Calibration of the one-tier tissue-transfer adjustment. A tissue-invariant nonsense/NMD effect reproduces more often than an RNA-route effect (87.0% versus 66.7% in the direct comparison). The difference corresponds to an odds ratio of 3.33, or 1.68 ACMG tiers, with a 95% confidence interval of -0.13 to 3.74 tiers.

Gene	Mechanism class	n	reproduces	reproduction rate	95% lo	95% hi
BRCA1	tissue-invariant (nonsense/NMD)	46	40	0.870	0.737	0.951
BRCA1	tissue-variable (RNA route)	105	70	0.667	0.568	0.756
VHL	tissue-variable (RNA route)	26	20	0.769	0.564	0.910

Table 11. Number of variants whose ACMG tier changes when the mechanism axis is included. All 611 changes are reductions for RNA-route losses. This comparison removes the mechanism component from the same pipeline rather than comparing with an unrelated method.

Gene	RNA-route losses downgraded
BARD1	172
BRCA1	64
PALB2	207
RAD51D	146
VHL	22
Total	611

Table 12. External evaluation on two one-channel SGE screens not used in method development. Agreement is measured against independent ClinVar variants with at least two review stars. Only the survival component is tested.

Gene	n benign	benign→Normal	benign agree	n path	path→LoF	path agree
RAD51C	331	325	0.982	109	109	1.000
DDX3X	28	28	1.000	63	50	0.794

Table 13. Final evidence codes for all 32,406 held-out variants. PS3 and BS3 represent protein-route evidence, PVS1_RNA represents RNA-route loss, and uncertain indicates that the method abstained.

Final evidence code	Held-out variants
BS3_Strong	17353
BS3_Moderate	8465
PS3_Strong	2774
uncertain	2273
PS3_Moderate	475
PVS1_RNA_Moderate	448
PVS1_RNA_Supporting	356
PS3_Supporting	262

Table 14. Precision of the RNA-drop flag against independent SpliceVarDB experiments using the preliminary and Wilks-certified thresholds. Recovery is unchanged: both thresholds identify the same 16 confirmed splice-altering variants, including the same numbers missed by SpliceAI.

Threshold set	Candidates	SpliceVarDB overlap	Confirmed splice-altering	Conservative precision	Inclusive precision	SpliceAI misses @0.2/0.5/0.8
Naive 5%-FPR cut	1614	33	16	0.842	0.906	1 / 7 / 14
Error-bounded (Wilks)	1304	25	16	0.889	0.920	1 / 7 / 14